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Science and Civilization

IT IS IN TIMES of economic distress such as we experience everywhere today, one sees very clearly the strength of the moral forces that live in a people. Let us hope that a historian delivering judgment in some future period when Europe is politically and economically united, will be able to say that in our days the liberty and honour of this Continent was saved by its Western nations, which stood fast in hard times against the temptations of hatred and oppression; and that Western Europe defended successfully the liberty of the individual which has brought us every advance of knowledge and invention—liberty without which life to a self-respecting man is not worth living.

It cannot be my task today to act as judge of the conduct of a nation which for many years has considered me as her own; perhaps it is an idle task to judge in times when action counts.

Today, the questions which concern us are: how can we save mankind and its spiritual acquisitions of which we are the heirs? How can one save Europe from a new disaster?

It cannot be doubted that the world crisis and the suffering and privations of the people resulting from the crisis are in some measure responsible for the dangerous upheavals of which we are the witness. In such periods discontent breeds hatred, and hatred leads to acts of violence and revolution, and often even to war. Thus distress and evil produce new distress and new evil. Again the leading statesmen are burdened with tremendous responsibilities just the same as twenty years ago. May they succeed through timely agreement to establish a condition of unity and clarity of international obligations in Europe, so that for every State a warlike adventure must appear as utterly hopeless. But the work of statesmen can succeed only if they are backed by the serious and determined will of the people.

We are concerned not merely with the technical problem of securing and maintaining peace, but also with the important task of education and enlightenment. If we want to resist the powers which threaten to suppress intellectual and individual freedom we must keep clearly before us what is at stake, and what we owe to that freedom which our ancestors have won for us after hard struggles.

Without such freedom there would have been no Shakespeare, no Goethe, no Newton, no Faraday, no Pasteur and no Lister. There would be no comfortable houses for the mass of the people, no railway, no wireless, no protection against epidemics, no cheap books, no culture and no enjoyment of art for all. There would be no machines to relieve the people from the arduous labor needed for the production of the essential necessities of life. Most people would lead a dull life of slavery just as under the ancient despotisms of Asia. It *is* only men who are free, who create the inventions and intellectual works which to us moderns make life worth while.

Without doubt the present economic difficulties will eventually bring us to the point where the balance between supply of labor and demand of labor, between production and consumption, will be enforced by law. But even this problem we shall solve as free men and we shall not allow ourselves for its sake to be driven into a slavery, which ultimately would bring with it stagnation of every healthy development.

In this connection I should like to give expression to an idea which has occurred to me recently. I lived in solitude in the country and noticed how the monotony of a quiet life stimulates the creative mind. There are certain callings in our modern organization which entail such an isolated life without making a great claim on bodily and intellectual effort. I think of such occupations as the service in lighthouses and lightships. Would it not be possible to fill such places with young people who wish to think out scientific problems, especially of a mathematical or philosophical nature? Very few of such people have the opportunity during the most productive period of their lives to devote themselves undisturbed for any length of time to scientific problems. Even if a young person is lucky enough to obtain a scholarship for a short period he must endeavor to arrive as quickly as possible at definite conclusions. That cannot be of advantage in the pursuit of pure science. The young scientist who carries on an ordinary practical profession which maintains him is in a much better position—assuming of course that this profession leaves him with sufficient spare time and energy. In this way perhaps a greater number of creative individuals could be given an opportunity for mental development than is